The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

Anonymous Submission

***Abstract*—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels within limited model and tool budgets. The Verified-Candidate Loop (VCL) lets validation evidence control candidate promotion. Its quality-of-results (QoR) score combines effective latency and worst-resource growth. QoR-guided retrieval-augmented generation (QoR-RAG) reuses compatible evidence, and Failure Reflection turns tool diagnostics into the next constraint. We evaluate three hosted large language model endpoints on a balanced 150-task suite. Qwen3.5 reaches 96.0% audit-clean success and the highest score sum, 7850.21. DeepSeek reaches the highest mean over tasks with an available score, 71.95, while Qwen3.6 reaches 48.7% success. Unequal provider failures limit model-only conclusions, so the comparison measures deployed endpoints.**

***Index Terms*—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results**

I. Evidence Must Control LLM Edits

Large language model (LLM) agents can edit unfamiliar high-level synthesis (HLS) kernels, but only tool reports certify correctness and hardware quality. A metered agent therefore needs a promotion rule that preserves a verified fallback when an edit fails simulation, synthesis, timing, capacity, or co-simulation [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents also alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

Our Verified-Candidate Loop (VCL) contributes three artifacts and findings. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across interface, simulation, synthesis, frequency, capacity, and required co-simulation checks. **(2) Hardware objective.** The bounded score Q_{HW} rejects speedups that lose more clock or worst-resource quality than they gain in latency. **(3) Endpoint evaluation.** Across 450 task runs, Qwen3.5 reaches 96.0% audit-clean success and the highest score sum, 7850.21. DeepSeek leads the scored-task mean at 71.95. Unequal application programming interface (API) failures limit this comparison to the deployed endpoints; matched ablations remain future work.

II. Tool Evidence Controls the Agent Workflow

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding hidden tests and frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs interface checks, C simulation (CSim), HLS synthesis (Synth), the 100 MHz frequency check, the U55C capacity check, task-required C/register-transfer-level (RTL) co-simulation (CoSim), and metric-completeness checks. For candidate c, define validity $V(c)$ over applicable gates $G_i \in \{0,1\}$; an inapplicable CoSim gate contributes one:

$$V(c) = G_{\mathrm{int}} G_{\mathrm{csim}} G_{\mathrm{syn}} G_{\mathrm{freq}} G_{\mathrm{cap}} G_{\mathrm{cosim}} G_{\mathrm{metric}}. \tag{1}$$

Let b denote the verified fallback. The promotion rule is

$$P(c) = V(c) \wedge [Q_{\mathrm{HW}}(c) > Q_{\mathrm{HW}}(b)]. \tag{2}$$

Budget admission reserves the credits for all required gates. A failed gate or budget denial retains b. QoR-RAG supplies version-compatible rules, successes, and measured failures. Failure Reflection converts bounded diagnostics into the next constraint without another LLM request. Duplicate candidates and repeated failures stop retries. Hidden acceptance remains an evaluator-side check.

Figure 1 summarizes the loop. The three campaigns record 573 candidate interface checks, 86 interface rejections, 409 synthesis gates, and 63 CoSim gates. The audit establishes contract execution but does not isolate mechanism effects.

III. One Score Rejects False Speedups

The evaluator selects a valid starter as anchor a and falls back to a frozen reference hidden from the agent. Let c denote a candidate, L its cycle latency, $p = \max(p_{\mathrm{target}}, p_{\mathrm{estimated}})$ its effective period, and II its initiation interval. The score uses the II branch only when the task marks II relevant and both values are positive:

$$r_L = \frac{p_a L_a}{p_c L_c}, \quad r_P = \begin{cases} r_L^{0.85} (II_a / II_c)^{0.15}, & \text{reliable } II, \\ r_L, & \text{latency only}, \end{cases}. \tag{3}$$

For anchor and candidate resource counts A_j and C_j, let $\mathcal{R}$ contain lookup tables (LUTs), flip-flops (FFs), digital signal processing (DSP) blocks, block RAM (BRAM), and UltraRAM (URAM). The worst growth defines area quality:

$$\begin{aligned} \mathcal{R}_+ &= \{j \in \mathcal{R} : A_j > 1 \vee C_j > 1\}, \\ \widehat{\mathcal{R}} &= \begin{cases} \mathcal{R}_+, & \mathcal{R}_+ \neq \varnothing, \\ \mathcal{R}, & \mathcal{R}_+ = \varnothing, \end{cases} \\ r_A &= \left[\max_{j \in \widehat{\mathcal{R}}} \frac{\max(C_j, 1)}{\max(A_j, 1)} \right]^{-1}. \end{aligned} \tag{4}$$

Task and budget →	Candidate and evidence →	Validation chain	→ Promote or retain
Kernel and interface →	LLM + QoR-RAG	→ CSIM/SYNTH/frequency/capacity/COSIM →	Verified fallback
→	← Failure Reflection from measured diagnostics		

Fig. 1. **Measured evidence controls every state transition.** A candidate replaces the verified fallback only after every applicable gate passes and Q_{HW} improves.

The evaluator score combines validity, bounded hardware utility, and budget efficiency:

$$h = r_P^{0.55} r_A^{0.45}, \qquad Q_{\mathrm{HW}} = 1 - \frac{1}{(1+h)^2}, \tag{5}$$

$$E = \max\big(0.80,\, 1 - 0.10u_{\mathrm{credit}} - 0.10u_{\mathrm{time}}\big), \quad S = 100 V Q_{\mathrm{HW}} E. \tag{6}$$

Here the two u terms denote consumed fractions clipped to $[0, 1]$. Promotion compares Q_{HW} only after $V(c) = 1$, so remaining budget never turns an inferior design into the fallback.

IV. Qwen3.5 Leads This Endpoint Run Set

Protocol. Three hosted endpoints process the same balanced 150-task suite under one agent revision and 9,000-credit limit. Each completed output must pass CSim, Synth, 100 MHz, U55C capacity, and every applicable CoSim gate under the Track-A contract [8]. Audit-clean success also excludes any task with a failed API request. Appendix A gives task provenance; Appendix C gives score and cost details.

TABLE I
Qwen3.5 combines the highest audit-clean success and score sum. Tokens are in millions; "API tasks" had at least one failed request. Score sum includes every non-null evaluator score, including scores from tasks rejected by the API audit.

Endpoint	Audit-clean	Completed	Score sum	Tokens	API tasks
DeepSeek V4 Pro	122/150	134/150	7483.07	2.72	25
Qwen3.5-122B-A10B	144/150	146/150	7850.21	4.72	2
Qwen3.6-27B	73/150	86/150	4482.31	2.57	66

Results. Qwen3.5 records 144/150 audit-clean successes and the highest score sum, 7850.21. DeepSeek completes 134 tasks but passes the API audit on 122; it still has the highest scored-task mean, 71.95. Qwen3.6 encounters failed requests on 66 tasks, and 53 tasks fail without a response. Conditional means cover different subsets, while unequal provider failures confound model-weight comparisons.

Takeaway. Qwen3.5 provides the strongest deployed endpoint in this run set. DeepSeek's higher conditional mean still yields a lower aggregate score.

V. Limits and Next Steps

The VCL makes validation evidence and Q_{HW} the promotion authority. Across 450 runs, Qwen3.5 combines 96.0% audit-clean success with the highest score sum, 7850.21. One campaign per endpoint conflates model effects, provider failures, and sampling variation. Matched ablations must establish the causal effects of QoR-RAG and Failure Reflection. Broader repositories, tool versions, and devices must test transfer beyond the present AMD corpus.

References

[1] Advanced Micro Devices, Inc., *Vitis High-Level Synthesis User Guide*, AMD, 2026, version 2025.2. [Online]. Available: https://docs.amd.com/r/2025.2-English/ug1399-vitis-hls

[2] C. Xiong, C. Liu, H. Li, and X. Li, "HLSPilot: LLM-based high-level synthesis," in *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design*, 2024, pp. 1–9.

[3] H. Xu, H. Hu, and S. Huang, "Optimizing high-level synthesis designs with retrieval-augmented large language models," in *2024 IEEE LLM Aided Design Workshop*, 2024, pp. 1–5.

[4] R. Li, J. Xiong, X. He, J. Zhao, J. Lv, H. Fang, L. Qi, and X. Wang, "ChatHLS: Towards systematic design automation and optimization for high-level synthesis," *arXiv preprint arXiv:2507.00642*, 2026. [Online]. Available: https://arxiv.org/abs/2507.00642

[5] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations*, 2023.

[6] Advanced Micro Devices, Inc., "Vitis-HLS-Introductory-Examples," https://github.com/Xilinx/Vitis-HLS-Introductory-Examples, 2026, apache-2.0.

[7] ——, "Vitis Acceleration Examples," https://github.com/Xilinx/Vitis_Accel_Examples, 2026, mIT.

[8] FPT 2026 Organizing Committee, "FPT'26 Design Competition: Track A," https://fpt2026.uark.edu/fpt26-design-competition/, 2026, accessed July 2026.

APPENDIX A
BENCHMARK SOURCES AND TASK COMPOSITION

The accepted manifest freezes every upstream path, commit, task description, public test, evaluator-only hidden test, reference, and gate configuration. The builder rejected construction candidates that failed acceptance checks and retained 150 tasks with no cross-category kernel overlap. Table II separates six capabilities and exposes weak repair modes hidden by an aggregate success rate.

The kernels come from two public AMD repositories at fixed commits. The introductory examples contribute 109 tasks under Apache-2.0; the acceleration examples contribute 41 under MIT. Controlled faults define generation and repair conditions, while frozen valid starters or evaluator-only references anchor QoR evaluation. The 150 variants reuse 65 unique source paths. The source repositories are https://github.com/Xilinx/Vitis-HLS-Introductory-Examples and https://github.com/Xilinx/Vitis_Accel_Examples.

APPENDIX B
TRACK-A REQUIREMENT MAPPING AND SCOPE

Table IV distinguishes runtime-output evidence from submission-package obligations. "Strict" refers to fail-closed enforcement: missing evidence for an applicable gate rejects the output. Strict enforcement allows inapplicable gates to remain unexecuted. All tasks require CSIM and SYNTH, whereas the manifest requires COSIM for the 25 structural tasks. The public guideline also says to pass COSIM without defining task-level exemptions. This report therefore scopes its COSIM evidence to the structural tasks and does not claim COSIM coverage for the remaining 125 tasks. Organizer confirmation or broader evidence must close this submission requirement.

APPENDIX C
DETAILED THREE-MODEL RESULTS

All campaigns cover the same 150 tasks and share one source hash and credit limit. Table VI reports audit-clean acceptance and optimization quality. The all-task mean assigns zero to a task without an evaluator score; the scored mean excludes tasks without a valid QoR anchor. A scored task can still fail the audit-clean success gate.

DeepSeek fails 28 tasks, Qwen3.5 fails 6, and Qwen3.6 fails 77. Failed API requests affect 25, 2, and 66 tasks, respectively. DeepSeek has 12 failed tasks with no API response; Qwen3.6 has 53, including every compile and functional task. Qwen3.5 has none. Failed requests cause the strict API audit to reject 12, 2, and 13 otherwise completed runs. We therefore attribute the tables to hosted endpoints and avoid a model-weights ranking. The evidence leaves counterfactual performance under a healthy provider unknown for starved tasks.

Qwen3.6's lower token total and shorter wall span partly reflect early request failure, so they do not establish higher efficiency. Table X also avoids retry-independence assumptions: its HTTP rate describes requests, not the probability that a complete model call fails.

Required ablations. A matched rerun should add one row per agent variant and report campaign identifier, success, repeated failures, mean Q_{HW}, tokens, calls, credits, and wall time. Planned variants are full agent, no QoR-RAG, no measured-failure retrieval, and no Failure Reflection. Until such runs exist, we print no value and claim no causal improvement.

APPENDIX D
REPRODUCTION AND RESULT REFRESH

The accepted task list is `tasks/track_a_150/candidate_manifest.json`. The three source report paths appear as comments at the end of `technical-paper/results_generated.tex`. After replacing or extending matching directories under `runs/`, execute `python3 technical-paper/scripts/update_results.py` from the repository root. The script selects the latest matching final report and refuses to emit macros unless model identity, task count, API client, execution-source hash, and credit limit agree.

The generated file supplies task counts, result prose values, and all detailed table rows. A refreshed campaign updates numerical evidence without hand-editing the paper. Evaluator-only tests and references remain outside the agent-visible filesystem; report writers redact credentials and sensitive paths.

The campaigns use source snapshot `0a06af39777b6ae7f3962afa2910232eaf782e91727e0f184ec168f1a41f106c`, temperature 0, a 4,096-token output limit, a 180-second request timeout, and up to two retries after the first request. OpenRouter identifies the requested open-weight model and license mapping. The opaque provider build remains unfrozen. The run directory names retain campaign timestamps.

TABLE II

THE TEST SUITE BALANCES SIX AGENT CAPABILITIES. EACH CATEGORY CONTAINS 25 TASKS; ONLY THE STRUCTURAL CATEGORY REQUIRES C/RTL CO-SIMULATION IN THE ACCEPTED MANIFEST.

Task type	Initial condition	Required output	Tasks	CoSim
Code generation	Signature-only stub	Complete kernel	25	0
Compile repair	Compile failure	Compiling kernel	25	0
Synthesis repair	Synthesis failure	Synthesizable kernel	25	0
Functional repair	CSim failure	Functionally correct kernel	25	0
Structural repair	CoSim failure	Executable RTL structure	25	25
QoR optimization	Valid baseline	Better valid Q_{HW}	25	0

TABLE III

FIXED SOURCE REVISIONS MAKE THE CORPUS RECONSTRUCTABLE.

Repository	License	Upstream commit	Tasks
Xilinx/Vitis-HLS-Introductory-Examples	Apache-2.0	`aa5c160faf5d5ebf58674df8f0591f9984ebae0f`	109
Xilinx/Vitis_Accel_Examples	MIT	`81187602355a7c2b666351154c5acca2074cae64`	41

TABLE IV

THE RUN CONTRACT MAPS THE APPLICABLE TRACK-A EVALUATION REQUIREMENTS TO RECORDED EVIDENCE. SUBMISSION FILES AND VIDEO REMAIN AUTHOR-SIDE CHECKLIST ITEMS.

Track-A requirement	Enforcement and recorded evidence
U55C; Vitis 2025.2	The container fixes the target part and tool release; reports preserve the execution-source revision.
CSim, Synth, applicable CoSim	A missing or failed required gate rejects the candidate. Hidden tests and frozen references remain evaluator-only.
At least 100 MHz; feasible resources	Frequency and U55C capacity gates precede promotion. Reports retain latency, II, period, LUT, FF, DSP, BRAM, and URAM.
Budget and token accounting	Every report records prompt, completion, and total tokens, API requests, tool calls, credits, and wall time. The campaign limit is 9,000 credits.
Recommended model comparison	OpenRouter campaigns cover DeepSeek V4 Pro, Qwen3.5-122B-A10B, and Qwen3.6-27B under one manifest and source revision. Mock, replay, and scripted backends cause the result generator to stop.
Docker and reproducibility	The repository supplies the Docker workflow, public task materials, structured reports, and result-generation script.
PDF, source archive, video	Authors separately confirm IEEE format, the two-page boundary, archive contents, and the five-minute demonstration as package obligations.

TABLE V

THE PUBLIC GUIDELINE IDENTIFIES THREE RECOMMENDED CHECKPOINTS. ARCHITECTURE AND QUANTIZATION DESCRIBE THE LISTED CHECKPOINTS; THE HOSTED PROVIDER BUILD REMAINS OPAQUE. MOE MEANS MIXTURE-OF-EXPERTS, FP8 MEANS 8-BIT FLOATING POINT, AND AWQ MEANS ACTIVATION-AWARE WEIGHT QUANTIZATION.

Endpoint	Architecture	Total/active parameters	Quantization	Context
DeepSeek V4 Pro	MoE	1.6T/49B	FP8	1,000,000
Qwen3.5-122B-A10B	MoE	122B/10B	AWQ-4bit	262,144
Qwen3.6-27B	Dense	27B/27B	FP8	262,144

TABLE VI

AGGREGATE OUTCOMES SEPARATE ACCEPTANCE, OPTIMIZATION QUALITY, AND COVERAGE.

Model	Audit-clean success	Scored tasks	Scored mean	All-task mean	Score sum
DeepSeek V4 Pro	122/150	104	71.95	49.89	7483.07
Qwen3.5-122B-A10B	144/150	115	68.26	52.33	7850.21
Qwen3.6-27B	73/150	67	66.90	29.88	4482.31

TABLE VII

QoR tasks expose normalized performance and resource quality. Each campaign evaluates the same 25 QoR tasks. Their evaluator score combines normalized performance r_P, worst-resource quality r_A, validity, and budget use; the frozen evaluator does not report power, and raw cross-kernel resource counts are not averaged.

Model	Audit-clean QoR success	QoR score mean (all 25)
DeepSeek V4 Pro	24/25	80.49
Qwen3.5-122B-A10B	24/25	76.55
Qwen3.6-27B	15/25	76.43

TABLE VIII

Endpoint availability shapes the category spread. Entries give audit-clean successes out of 25 tasks and the corresponding rate.

Model	Generate	Compile	Synth	Functional	Structural	QoR
DeepSeek V4 Pro	24/25 (96%)	25/25 (100%)	17/25 (68%)	18/25 (72%)	14/25 (56%)	24/25 (96%)
Qwen3.5-122B-A10B	22/25 (88%)	25/25 (100%)	25/25 (100%)	23/25 (92%)	25/25 (100%)	24/25 (96%)
Qwen3.6-27B	15/25 (60%)	0/25 (0%)	25/25 (100%)	0/25 (0%)	18/25 (72%)	15/25 (60%)

TABLE IX

The reports retain model and tool budget costs. Token counts are in millions; API failures count requests without a usable response.

Model	Prompt	Completion	Total	API requests	Responses	API failures	Credits	Tool calls
DeepSeek V4 Pro	1.78	0.94	2.72	447	346	101	2,398	649
Qwen3.5-122B-A10B	1.75	2.96	4.72	335	333	2	2,650	719
Qwen3.6-27B	1.55	1.02	2.57	636	282	354	1,805	466

TABLE X

Derived costs use audit-clean success as the denominator. HTTP failures count raw requests and may cluster within one logical model call. "Wall span" measures the parallel campaign span.

Model	Tokens/success	Requests/success	Credits/success	HTTP failure rate	Wall span (h)
DeepSeek V4 Pro	22,299.9	3.66	19.66	22.6%	4.54
Qwen3.5-122B-A10B	32,747.7	2.33	18.40	0.6%	4.94
Qwen3.6-27B	35,138.9	8.71	24.73	55.7%	3.46

TABLE XI

Tool calls expose validation effort under the shared budget.

Model	CSim	Synth	CoSim	Total
DeepSeek V4 Pro	306	298	45	649
Qwen3.5-122B-A10B	342	327	50	719
Qwen3.6-27B	249	174	43	466

TABLE XII

Task-level evidence separates clean success from API exposure. "Completed" records the agent run outcome before the strict API audit. "API affected" counts tasks with at least one failed request.

Model	Audit-clean success	Completed	API affected	Affected but completed	Failed with zero responses
DeepSeek V4 Pro	122	134	25	12	12
Qwen3.5-122B-A10B	144	146	2	2	0
Qwen3.6-27B	73	86	66	13	53